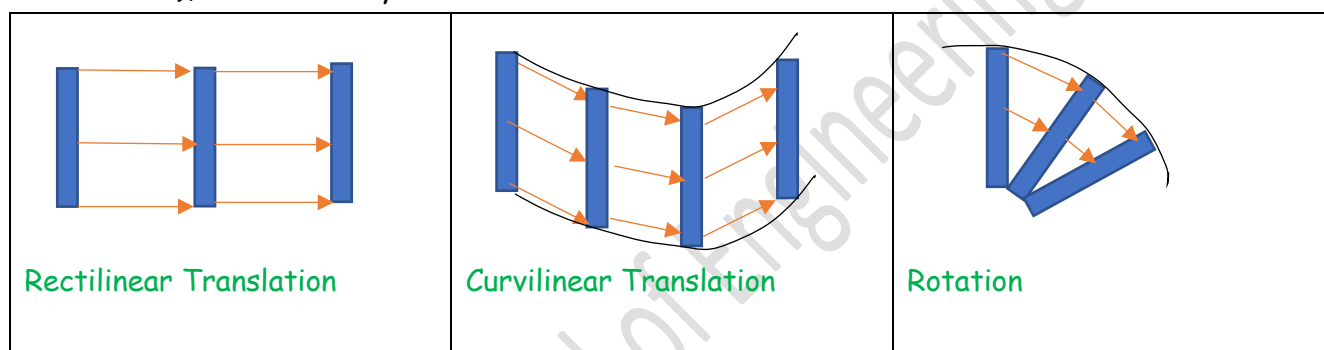


II. DYNAMICS

We know when forces act on any system, the system may remain in equilibrium or it may move. So far, we have been discussing the force systems in equilibrium. Now let us discuss the dynamics. The equations of static equilibrium for a force system in a plane are $\sum F_x = 0$, $\sum F_y = 0$, $\sum M_z = 0$. If these conditions are not satisfied, the body will start moving. Then we are dealing with a dynamic system.

$\sum F_x \neq 0$, $\sum F_y = 0$, $\sum M_z = 0$ Rectilinear Translation	$\sum F_x \neq 0$, $\sum F_y \neq 0$, $\sum M_z = 0$ Curvilinear Translation	$\sum F_x = 0$, $\sum F_y = 0$, $\sum M_z \neq 0$ Rotation	$\sum F_x \neq 0$, $\sum F_y = 0$, $\sum M_z \neq 0$ Plane motion
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When a body moves in such a way that the displacement, velocity, and acceleration of any point on that body are the same at any instant of time, we consider the motion as a **translation**. The body can move along a straight line (**Rectilinear Translation**) or along some curve (**curvilinear translation**). When the different points of the body have different displacements (or velocity, acceleration), then the body is said to have a **rotation**.



When a rigid body is translating, only its **mass** is important and not its mass distribution. Since all mass points in that body have the same motion parameters, we need to consider only one point in that body. Usually that point is the **centre of mass**, and we consider the body as a **particle**. This is not possible when a rigid body is rotating. Since, different mass points have different motion parameters, we need to consider not only the mass of the body but its distribution with respect to the axis of rotation. Hence when a body is rotating, its **mass moment of inertia** becomes important.

When we deal with bodies in motion, if we consider only the motion parameters like displacement, velocity, acceleration, time, path etc. without referring to the forces that cause motion, we are referring to the **kinematics** of motion. If we consider the forces that cause the motion, we are dealing with the **kinetics** of motion.

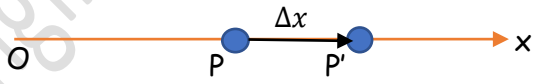
KINEMATICS OF RECTILINEAR TRANSLATION

Displacement

When a body moves along a straight line, the line of motion is considered as the x-axis. The position of the body is a function of time, $x(t) = f(t)$. As the time changes, the position also changes. Consider a body whose position at time t_1 with respect to an origin is x_1 . When the time is t_2 , let the position changes to x_2 . That is, in time $\Delta t (= t_2 - t_1)$, the position changes by $\Delta x (= x_2 - x_1)$. The change in the position of the body is known as **displacement**.

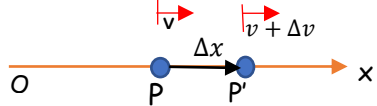
Velocity

The rate of change of displacement is known as the velocity. If the body changes its position by Δx in a time Δt , its velocity is defined as $\frac{\Delta x}{\Delta t}$. If in equal intervals of time Δt , the body has equal displacements; the body is said to have a **uniform velocity**. If in equal intervals of time, the displacements are not equal, we can find an average velocity by $\frac{\Delta x}{\Delta t}$, i.e., by dividing the total displacement by the total elapsed time. In such a situation, we divide the total time in to infinitesimally small intervals (Δt). Let us consider the body be at P at any instant of time t . In the next instant of time $t + \Delta t$, the body moves to P' so that the displacement $PP' = \Delta x$. Then $\lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$ will give the velocity of the body at the instant from which we have considered Δt . It is known as the **instantaneous velocity**.

i.e., the instantaneous velocity, $v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} = \dot{x}$ $v = \frac{dx}{dt} \rightarrow dx = v \cdot dt$, the displacement of the body from time t_1 to t_2 is given by $\int_{x_1}^{x_2} dx = \int_{t_1}^{t_2} v dt$ (Area of velocity-time diagram) If v is constant (uniform), $x_2 - x_1 = x = v(t_2 - t_1)$	
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Acceleration

Acceleration is the rate of change of velocity. If a body changes its velocity by Δv in time Δt , its acceleration is defined as $\frac{\Delta v}{\Delta t}$. If in equal intervals of time Δt , the body has equal change in velocity; the body is said to have a **uniform acceleration**. If in equal intervals of time, the changes in velocity are not equal, we can find an average acceleration by $\frac{\Delta v}{\Delta t}$, i.e., by dividing the total change in velocity by the total elapsed time. Let us divide the total time in to infinitesimally small intervals (Δt). Consider that the body be at P at any instant of time t with a velocity v . In the next instant of time $t + \Delta t$, the body moves to P' with velocity $v + \Delta v$. Then $\lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$ will give the acceleration of the body at the instant from which we have considered Δt . It is known as the **instantaneous acceleration**.

i.e., the instantaneous acceleration, $a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \dot{v}$ Since $a = \frac{dv}{dt}$, or $dv = a dt$; the change in velocity of the body from time t_1 to t_2 is given by $\int_{v_1}^{v_2} dv = \int_{t_1}^{t_2} a dt$ $v_2 - v_1 = \int_{t_1}^{t_2} a dt$ (Area of acceleration-time diagram)	
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If a is constant(uniform),

$$v_2 - v_1 = a \int_{t_1}^{t_2} dt = a(t_2 - t_1)$$

If the body starts its motion with an initial velocity u and uniform acceleration a , its velocity v after a time t is given by

$$v = u + at$$

But $v = \frac{dx}{dt} \gg \gg \frac{dx}{dt} = u + at$

$$dx = (u + at)dt$$

Integrating, $\int_0^x dx = \int_0^t (u + at)dt$

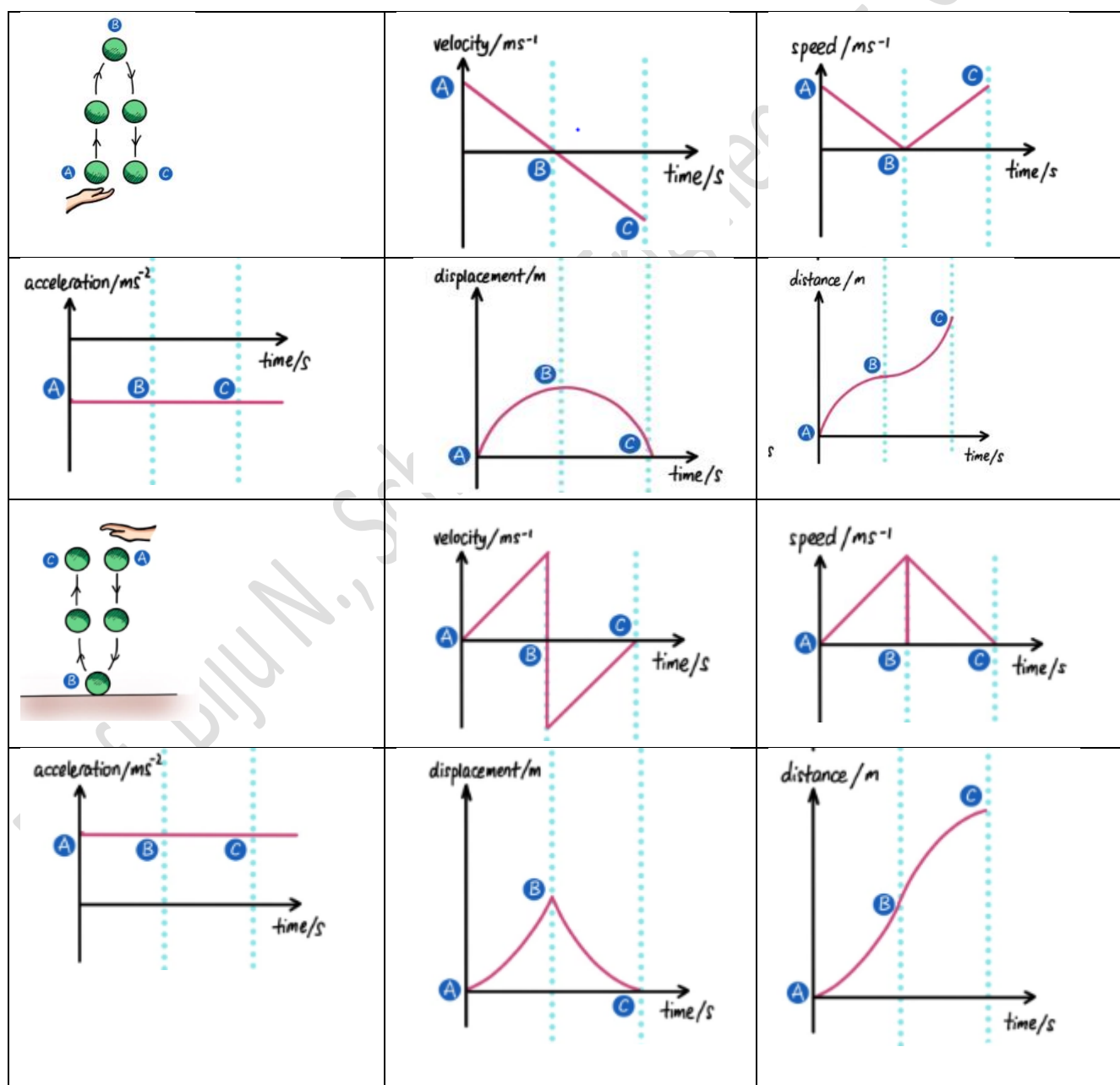
$$x = ut + \frac{1}{2}at^2$$

If velocity v is a function of displacement x , then acceleration $a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$

$$v dv = a dx$$

Integrating both sides $\int_u^v v dv = \int_0^x a dx \gg \gg \gg \frac{v^2 - u^2}{2} = ax$

$$v^2 = u^2 + 2ax$$



#Example 1. Consider a particle whose motion is described by the displacement-time equation $x = 6t^2 - t^3$. Find the velocity-time relation and acceleration-time relation. When will its velocity become maximum? Also find its velocity when its displacement becomes zero again.

Given, $x = 6t^2 - t^3$

Velocity, $v = \frac{dx}{dt} = 12t - 3t^2$

Acceleration, $a = \frac{dv}{dt} = 12 - 6t$

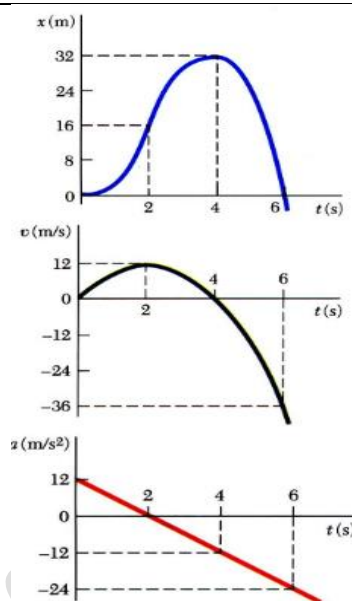
When v is maximum, $\frac{dv}{dt}$ (or acceleration) is zero.

$$12 - 6t = 0 \ggggg t = 2 \text{ s.}$$

When displacement is zero, $x = 6t^2 - t^3 = 0$

Or $t^2(6 - t) = 0$

$t = 0, 6 \text{ s}$



#Example 2. A car is moving in a straight line such that its velocity is defined by $v = 0.3(9t^2 + 2t)$ where v is in m/s and t is in seconds. Determine its position and acceleration when $t = 3 \text{ s}$, if it has no initial displacement.

Given the velocity-time relationship

$$v = 0.3(9t^2 + 2t)$$

The acceleration, $\ddot{x} = \frac{dv}{dt} = 0.3(18t + 2)$

When $t = 3 \text{ s}$, $\ddot{x} = 0.3(18t + 2) = 16.8 \text{ m/s}^2$

i.e., $v = \frac{dx}{dt} = 0.3(9t^2 + 2t)$

$$dx = 0.3(9t^2 + 2t)dt$$

Integrating both sides

$$\int_0^x dx = \int_0^3 0.3(9t^2 + 2t)dt$$

$$x = 0.3 \left(\frac{9t^3}{3} + \frac{2t^2}{2} \right) = 27 \text{ m}$$

#Example 3. The rectilinear motion of a particle is defined by the displacement-time equation $x = x_0(2e^{-kt} - e^{-2kt})$ where x_0 is the initial displacement and k is a constant. Find the maximum velocity of the particle.

Given, $x = x_0(2e^{-kt} - e^{-2kt})$

Velocity, $v = \frac{dx}{dt} = x_0(-2ke^{-kt} + 2ke^{-2kt})$

When velocity is maximum, $\frac{dv}{dt} = 0$

$$\frac{dv}{dt} = 2kx_0(ke^{-kt} - 2ke^{-2kt}) = 0$$

$$\frac{e^{-2kt}}{e^{-kt}} = \frac{1}{2} \gggg e^{-kt} = \frac{1}{2}$$

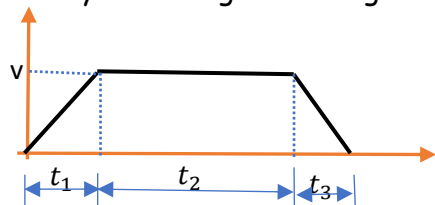
$$-kt \ln e = \ln\left(\frac{1}{2}\right) \gggggg t = \frac{\ln 2}{k}$$

The maximum velocity,

$$v_{\max} = x_0 \left(-2k \times \frac{1}{2} + 2k \times \frac{1}{4} \right) = \frac{-kx_0}{2}$$

Q 4. A train starting from rest increases its speed from 0 to v with a constant acceleration a_1 , runs at this speed for some time, and finally comes to rest with constant deceleration a_2 . If the total distance travelled is s , find the total time t elapsed.

The velocity-time diagram is as given below.



The motion can be divided into three.

i) Uniformly accelerated motion.

Let the train covers a distance of s_1 in time t_1 with a constant acceleration a_1

Initial velocity = 0

Final velocity = v

$$v = 0 + a_1 t_1 \gggg \gg t_1 = \frac{v}{a_1}$$

$$s_1 = 0 + \frac{1}{2} a_1 t_1^2 \gggg \gg s_1 = \frac{v^2}{2a_1}$$

ii) Uniform velocity motion

Let the train covers a distance of s_2 in time t_2 with a constant velocity v

$$s_2 = v t_2 \gggg \gg t_2 = \frac{s_2}{v}$$

iii) Uniform deceleration motion

Let the train comes to rest in time t_3 by covering a distance s_3 with constant deceleration a_2

Initial velocity = v

Final velocity = 0

$$0 = v - a_2 t_3 \gggg \gg t_3 = \frac{v}{a_2}$$

$$s_3 = v t_3 - \frac{1}{2} a_2 t_3^2 \gggg \gg s_3 = \frac{v^2}{2a_2}$$

$$s_2 = s - (s_1 + s_3) = s - \frac{v^2}{2a_1} - \frac{v^2}{2a_2}$$

Total time elapsed $t = t_1 + t_2 + t_3$

$$= \frac{v}{a_1} + \frac{s - \frac{v^2}{2a_1} - \frac{v^2}{2a_2}}{v} + \frac{v}{a_2} = \frac{s}{v} + \frac{v}{2a_1} + \frac{v}{2a_2}$$

$$t = \frac{s}{v} + \frac{v}{2} \left(\frac{1}{a_1} + \frac{1}{a_2} \right)$$

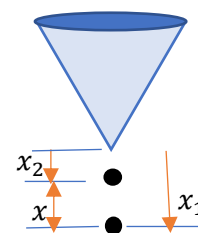
#Example 5. Water drips from a funnel at a rate of n drops per second. Neglecting air resistance, find the distance x between any two adjacent drops as a function of the time t that the trailing drop has been in motion.

Number of drops per second = n

Time gap between dripping of two successive drops = $\frac{1}{n}$ second

The drops fall under gravity. The initial velocity of the drops is zero and the acceleration, $a = g = 9.81 \text{ m/s}^2$.

Consider two successive drops. If t is the time that the second drop has been in motion, then the first drop has been in motion for $(t + \frac{1}{n})$ seconds.



The distance travelled by the first drop in time $(t + \frac{1}{n})$ seconds, $x_1 = 0 + \frac{1}{2} g (t + \frac{1}{n})^2$

The distance travelled by the second drop in time t seconds, $x_2 = 0 + \frac{1}{2} g (t)^2$

The distance between the drops, $x = x_1 - x_2 = \frac{1}{2} g \left(t^2 + \frac{2t}{n} + \frac{1}{n^2} \right) - \frac{1}{2} g (t)^2$

$$x = \frac{gt}{n} + \frac{g}{n^2}$$

#Exercise 28

28.1.

A stone is dropped in to a well and falls vertically with constant acceleration $g = 9.81 \text{ m/s}^2$. The sound of impact of the stone on the bottom of the well is heard 7 seconds after it is dropped. If the velocity of the sound in air is 340 m/s , how deep is the water surface?

28.2.

A train travels with a velocity $\dot{x} = 20(1 - e^{-t}) \text{ m/s}$. Where t is the elapsed time in seconds. Determine its acceleration and distance travelled after 3 s.

28.3.

A ball is fire downward in to water with an initial velocity of 27 m/s . If it experiences a deceleration $\ddot{x} = -6t \text{ m/s}^2$, where t is seconds. Determine the distance travelled before it stops.

28.4.

A ball is thrown vertically upward from origin under the influence of gravity with an initial velocity $\dot{x}_0 = 6 \text{ m/s}$. Find the maximum height to which it will rise and the time required for it to return to its initial position. Assume no air resistance.

28.5.

The maximum possible acceleration and deceleration that a car can have is a and its maximum speed is v . Find the minimum time in which the car travel from one city to another if the distance between them is s .

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